CHAPTER 8

Electromagnetic Waves

Displacement Current, Ampere Maxwell's Law

1. A parallel plate capacitor is charged by connecting it to a batter/through a resistor. If I is the current in the circuit, then in the gap between the plates: *(2024)*

a. displacement current of magnitude equal to I flows in a direction opposite to that of I.

b. displacement current of magnitude greater than I flows but can be in any direction.

c. there is no current.

d. displacement current of magnitude equal to I flows in the same direction as I.

2. An ac source is connected to a capacitor C. Due to decrease in its operating frequency: *(2023)*

a. capacitive reactance remains constant

b. capacitive reactance decreases.

c. displacement current increases.

d. displacement current decreases.

3. To produce an instantaneous displacement current of 2 mA in the space between the parallel plates of a capacitor of capacitor of capacitance 4μF, the rate of change of applied variable potential difference $\left(\frac{dV}{dt}\right)$ must be: *(2023-Manipur)*

a. 800 V/s b. 500 V/s c. 200 V/s d. 400 V/s

4. A 100 Ω resistance and a capacitor of 100 Ω reactance are connected in series across a 220 V source. When the capacitor is **50% charged**, the peak value of the displacement current is: *(2016 - II)*

a. 4.4 A b. $11\sqrt{2}$ A c. 2.2 A d. 11 A

Properties and Applications (i.e. Velocity, Amplitude, Energy Density) of Electromagnetic Waves

5. If the ratio of relative permeability and relative permittivity of a uniform medium is 1 : 4. The ratio of the magnitudes of electric field intensity (E) to the magnetic field intensity (H) of an EM wave propagating in that medium is

$\left(\text{Given that } \sqrt{\frac{\mu_0}{\varepsilon_0}} = 120\pi\right)$: *(2024 Re)*

a. 30π : 1 b. 1 : 120 π c. 60π : 1 d. 120 π : 1

6. The property which is not of an electromagnetic wave travelling in free space is that: *(2024)*

a. they travel with speed equal to $\frac{1}{\sqrt{\mu_0 \in_0}}$.

b. they originate from charges moving with uniform speed.

c. they are transverse in nature.

d. the energy density in electric field is equal to energy density in magnetic field.

7. In a plane electromagnetic wave travelling in free space, the electric field component oscillates sinusoidally at a frequency of 2.0×10^{10} Hz and amplitude 48 Vm^{-1}. Then the amplitude of oscillating magnetic field is:

(Speed of light in free space = 3 × 108 ms^{-1}) *(2023)*

a. 1.6×10^{-6} T b. 1.6×10^{-9} T

c. 1.6×10^{-8} T d. 1.6×10^{-7} T

8. The magnetic field of a plane electromagnetic wave is given by

$\vec{B} = 3 \times 10^{-5} \cos(1.6 \times 10^3 x + 48 \times 10^{10} t)\, \hat{j}$,

then the associated electric field will be: *(2022 Re)*

a. $9 \cos(1.6 \times 10^3 x + 48 \times 10^{10} t)\, \hat{k}$ V/m

b. $3 \times 10^{-8} \cos(1.6 \times 10^3 x + 48 \times 10^{10} t)\, \hat{i}$ V/m

c. $3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t)\, \hat{i}$ V/m

d. $9 \sin(1.6 \times 10^3 x - 48 \times 10^{10} t)\, \hat{k}$ V/m

9. The ratio of the magnitude of the magnetic field and electric field intensity of a plane electromagnetic wave in free space of permeability μ_0 and permittivity ε_0 is (Given that c - velocity of light in free space) *(2022 Re)*

a. $\frac{\sqrt{\mu_0 \varepsilon_0}}{c}$ b. c c. $\frac{1}{c}$ d. $\frac{c}{\sqrt{\mu_0 \varepsilon_0}}$

10. When light propagates through a material medium of relative permittivity $\in_r$ and relative permeability μ_r, the velocity of light, v is given by : (c - velocity of light in vacuum) *(2022)*

a. $v = \frac{c}{\sqrt{\in_r \mu_r}}$ b. v = c

c. $v = \sqrt{\frac{\mu_r}{\in_r}}$ d. $v = \sqrt{\frac{\in_r}{\mu_r}}$

11. For a plane electromagnetic wave propagating in x-direction, which one of the following combination gives the **correct** possible directions for electric field (E) and magnetic field (B) respectively? *(2021)*

a. $-\hat{j}+\hat{k}, -\hat{j}-\hat{k}$ b. $\hat{j}+\hat{k}, -\hat{j}-\hat{k}$

c. $-\hat{j}+\hat{k}, -\hat{j}+\hat{k}$ d. $\hat{j}+\hat{k}, \hat{j}+\hat{k}$

12. Light with an average flux of 20 W/cm^2 falls on non-reflecting surface at normal incidence having surface area 20 cm^2. The energy received by the surface during time span of **1 minute** is: *(2020)*

a. 12×10^3 J b. 24×10^3 J

c. 48×10^3 J d. 10×10^3 J

13. The ratio of contributions made by the electric field and magnetic field components to the **intensity** of an electromagnetic wave is : (c = speed of electromagnetic waves) *(2020)*

a. 1 : 1 b. 1 : c c. 1 : c^2 d. c : 1

14. The magnetic field in an electromagnetic wave is given by, $B_y = 2 \times 10^{-7} \sin(\pi \times 10^3 x + 3\pi \times 10^{11} t)$T

Calculate the wavelength. *(2020-Covid)*

a. 2×10^{-3} m b. 2×10^3 m

c. $\pi \times 10^{-3}$ m d. $\pi \times 10^3$ m

15. An em wave is propagating in a medium with a velocity $\vec{v} = \hat{i}v$. The instantaneous oscillating electric field of this em wave is along +y axis. Then the direction of oscillating magnetic field of the em wave will be along. *(2018)*

a. –y direction b. +z direction

c. –z direction d. –x direction

16. In an electromagnetic wave in free space the root mean square value of the electric field is E_{rms} = 6 V/m. The peak value of the magnetic field is: *(2017-Delhi)*

a. 2.83×10^{-8} T b. 0.70×10^{-8} T

c. 4.23×10^{-8} T d. 1.41×10^{-8} T

17. Out of the following options which one can be used to produce a propagating electromagnetic wave? *(2016 - I)*

a. A charge moving at constant velocity

b. A stationary charge

c. A charge less particle

d. An accelerating charge

18. Radiation of energy 'E' **falls normally** on a perfectly reflecting surface. The momentum transferred to the surface is (*C* = velocity of light): *(2015)*

a. $\frac{2E}{C}$ b. $\frac{2E}{C^2}$ c. $\frac{E}{C^2}$ d. $\frac{E}{C}$

19. Light with an energy flux of 25×10^4 W/m^2 falls on a perfectly reflecting surface at normal incidence. If the surface area is 15 cm^2, the average force exerted on the surface is: *(2014)*

a. 1.25×10^{-6} N b. 2.50×10^{-6} N

c. 1.20×10^{-6} N d. 3.0×10^{-6} N

Electromagnetic Spectrum

20. The electromagnetic radiation which has the smallest wavelength are *(2024 Re)*

a. X-rays b. Gamma rays

c. Ultraviolet rays d. Microwaves

21. Match List-I with List-II *(2022)*

List-I (Electromagnetic waves)	List-II (Wavelength)
A. AM radio waves	(i) 10^{-10} m
B. Microwaves	(ii) 10^2 m
C. Infra-red radiations	(iii) 10^{-2} m
D. X-rays	(iv) 10^{-4} m

Choose the correct answer from the options given below:

a. (A)-(ii), (B)-(iii), (C)-(iv), (D)-(i)

b. (A)-(iv), (B)-(iii), (C)-(ii), (D)-(i)

c. (A)-(iii), (B)-(ii), (C)-(i), (D)-(iv)

d. (A)-(iii), (B)-(iv), (C)-(ii), (D)-(i)

22. The E.M. wave with shortest wavelength among the following is, *(2020-Covid)*

a. X-rays b. Gamma-rays

c. Microwaves d. Ultraviolet rays

23. Which colour of the light has the longest wavelength? *(2019)*

a. Red b. Blue c. Green d. Violet

24. The energy of the E.M. waves is of the order of 15 keV. To which part of the spectrum does it belong? *(2015 Pre)*

a. Gamma-rays b. X-rays

c. Infra-red rays d. Ultraviolet rays

25. The condition under which a microwave oven heats up a food item containing water molecules most efficiently is: *(2013)*

a. Infra-red waves produce heating in a microwave oven

b. The frequency of the microwaves must match the resonant frequency of the water molecules

c. The frequency of the microwaves has no relation with natural frequency of water molecules

d. Microwaves are heat waves, so always produce heating

Effects of Dielectrics in Capacitors

26. When light propagates through a material medium of relative permittivity ϵ_r and relative permeability μ_r, the velocity of light, v is given by : (c - velocity of light in vacuum) *(2022)*

a. $v = \frac{c}{\sqrt{\epsilon_r \mu_r}}$ b. v = c

c. $v = \sqrt{\frac{\mu_r}{\epsilon_r}}$ d. $v = \sqrt{\frac{\epsilon_r}{\mu_r}}$

27. A parallel plate capacitor of capacitance 20 μF is being charged by a voltage source whose potential is changing at the rate of 3 V/s. The conduction current through the connecting wires, and the displacement current through the plates of the capacitor, would be, respectively. *(2019)*

a. Zero, 60 μA b. 60 μA, 60 μA

c. 60 μA, zero d. Zero, zero

Answer Key

1. (d)	**2.** (d)	**3.** (b)	**4.** (c)	**5.** (c)	**6.** (b)	**7.** (d)	**8.** (a)	**9.** (c)	**10.** (a)
11. (a)	**12.** (b)	**13.** (a)	**14.** (a)	**15.** (b)	**16.** (a)	**17.** (d)	**18.** (a)	**19.** (b)	**20.** (b)
21. (a)	**22.** (b)	**23.** (a)	**24.** (b)	**25.** (b)	**26.** (a)	**27.** (b)			